SAMARTH

CUET PG 2027

Lecture – 17

General LLB

Unit 3 : Digital Fundamental

Topic : Memory Unit

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Recap *of previous lecture*

1. Previous Year Questions + Doubt Class



Topics *to be covered*

1. Memory Unit

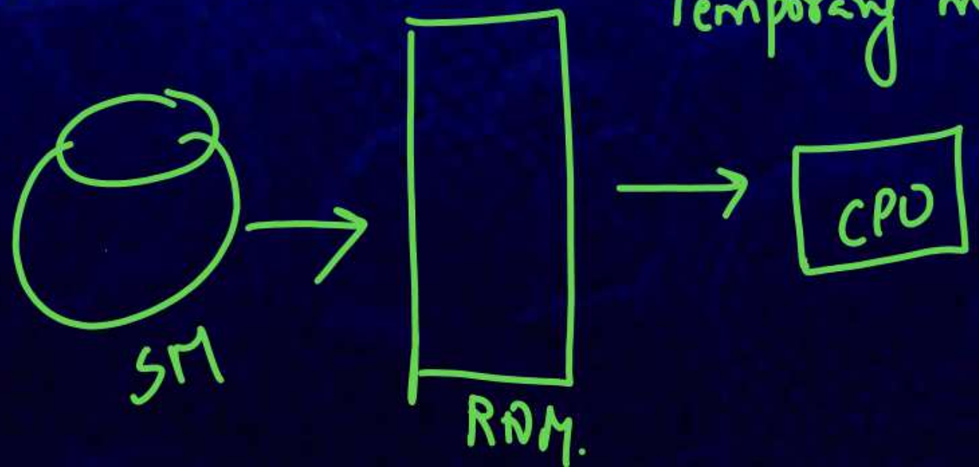


Topic : Memory Unit

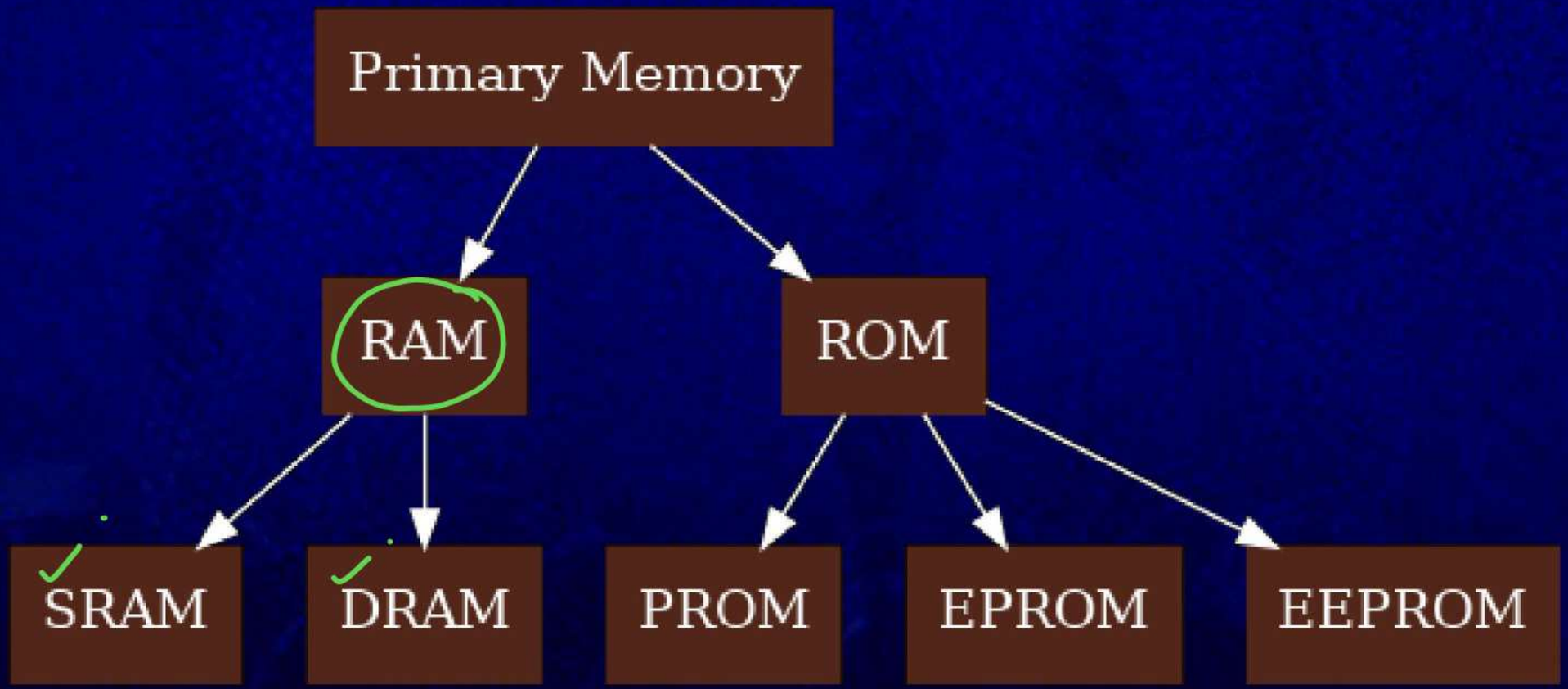
Topic : Memory Unit

Primary storage is also known as the main memory which includes RAM, ROM, Cache, registers. ***Primary Memory is volatile in nature.*** Data of primary memory may or may not vanish when power is lost depending on if it is stored in RAM or ROM.

- The data of ROM is non-volatile in nature, it is stored even when power is lost.
- The data of RAM is volatile in nature, it vanishes when power is lost.



Topic : Memory Unit



Topic : Memory Unit

1. ROM is a permanent memory which includes the data that need not to be stored. Anything stored in ROM cannot be changed so it is permanent memory.

Types of ROM:

- MROM: Masked ROM is hardwired and pre-programmed ROM. Content in MROM that is once written cannot be altered .
- PROM: Programmable ROM can be modified once by the user. A user buys a blank PROM and writes the required content however once written content cannot be altered.

Topic : Memory Unit

- EPROM: Erasable and Programmable ROM Content can be changed by erasing the initial content by exposing EPROM to UV radiation. This exposure to ultra-violet light dissipates the charge on ROM and content can be rewritten on it.
- EEPROM: Electrically Erasable and Programmable ROM Content can be changed by erasing electrically. only one byte can be erased at a time instead of deleting in one go.

Topic : Memory Unit

2. RAM:

- Dynamic RAM: Dynamic RAM or DRAM needs to periodically refresh in a few milliseconds to retain data. DRAM is made up of capacitors and transistors that needs to be charged periodically. DRAM is widely used in home PCs and servers as it is cheaper than SRAM.
- Static RAM: Static RAM or SRAM keeps the data as long as power is supplied to the system. It uses Sequential circuits like a flip-flop to store a bit and hence need not be periodically refreshed. SRAM is expensive and so only used where speed is the utmost priority.

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Differences between SRAM and DRAM:

SRAM	DRAM
It has lower access time, so it is faster compared to DRAM	It has higher access time, so it is slower than SRAM
It is costlier than DRAM	It costs less compared to SRAM
As it requires a constant power	It offers reduced power consumption due to the fact that the information is stored in the capacitor
Due to complex internal circuitry, less storage capacity is vailable compared to the same hysical size of the DRAM memory chip.	Due to small internal circuitry, large storage capacity is available
It has low storage density	It has a high storage density

Topic : Memory Unit

RAM VS ROM:

	ROM.
It stands for Random access memory.	**It stands for read only memory.**
It is the fastest memory.	It is slower memory as compare to RAM.
It is volatile memory. Temporary.	It is non-volatile memory. Permanent.
In this memory, data will erase when the power is off	In this memory, data will not erase even if the power is off

Topic : Memory Unit

Secondary Memory: It is auxiliary memory or backup memory. It is a non-volatile memory and used to store a large amount of data. The data or information stored in secondary memory is permanent, and it is slower than primary memory. CPU cannot access secondary memory directly. The data/information from the auxiliary memory is first transferred to the main memory, and then the CPU can access it. There are various types of secondary memories:

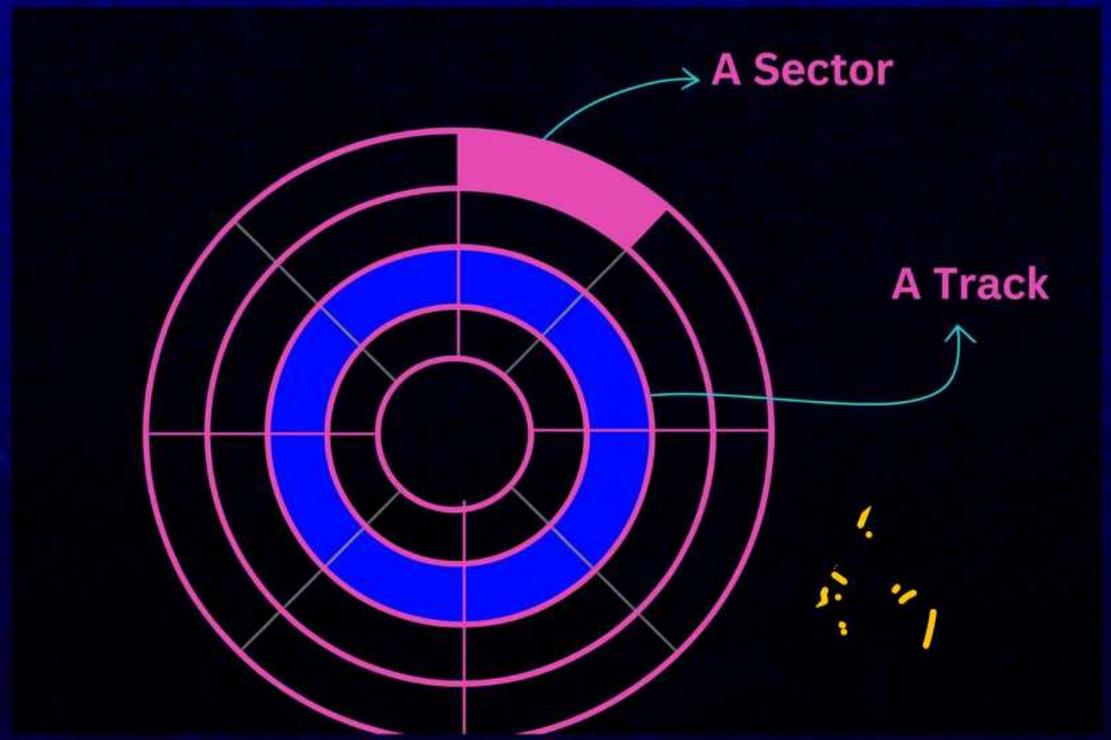
1. Magnetic tapes: It is a long, narrow strip of plastic film with a thin, magnetic coating on it that is used for magnetic recording. Bits are recorded on tape as magnetic patches called RECORDS that run along many tracks. It can be stopped, started moving forward or backward, or rewound.

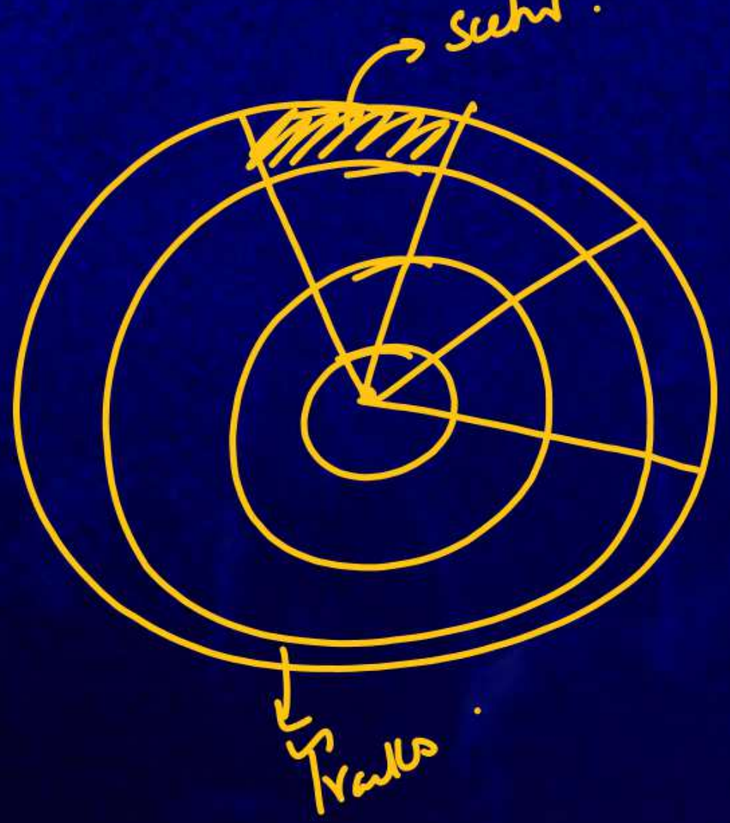
Topic : Memory Unit

2. Magnetic disk: is a circular metal or a plastic plate and these plates are coated with magnetic material. Bits are stored in magnetized surfaces in locations called tracks that run in concentric rings. Sectors are typically used to break tracks into pieces.

Hard disk

OS





Topic : Memory Unit

3. Optical disk: It's a laser-based storage medium that can be written to and read. It is reasonably priced and has a long lifespan.

a. CD-ROM: Information is written to the disc by using a controlled laser beam to burn pits on the disc surface. Its capacity is 600 MB, with each sector storing 2048 bytes of data. The data transfer rate is about 4800KB/sec. & the new access time is around 80 milliseconds.

b. WORMWRITE ONCE READ MANY: information is written on the disc using a laser beam. It is possible to read the written data as many times as desired. It has capacity of 5.25 inch disk is 650 MB,5.2GB etc

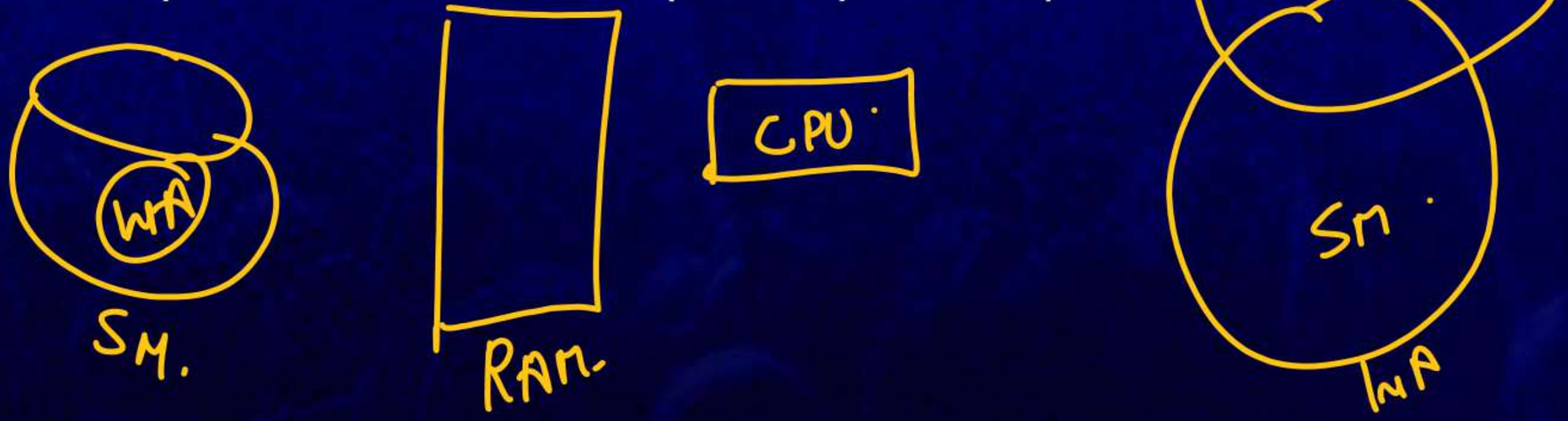
c. DVD: It is "Digital Versatile/Video Disc,". It is of two types DVDR (writable) and DVDRW(Re-writable).

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It is a concept.

- Virtual Memory :- It is virtual resource of a computer. It is an illusion that a computer system processes more memory that is actually having. This illusion makes a process independent of the size of real memory (main memory). If also permits a large number of processes to share computer system without constraining each other. In other words, virtual memory allows execution of partiallyloaded processes.

Details in OS



Topic : Memory Unit

- As a consequence, the sum of virtual address space of active processes in a virtual-memory space of active processes in a virtual-memory system can be exceed the capacity of the available physical memory provided that the physical memory is large enough to hold a minimum amount of address space of each active process. The choice of whichsectionsto bring in, when to bring them in and where to place them is made by the operating system.
- Thus, virtual memory system provide for automatic migration of portions of address space between secondary and primary storage of memory.

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→ Frequently executed data.

- Cache memory: It is a type of high-speed semiconductor memory that can help the CPU run faster. Between the CPU and the main memory, it serves as a buffer. It is used to store the data and programs that the CPU uses the most frequently. compared to the main memory, it takes less time to access it. It is very expensive and small in size.

SRAM

Speed ↑
SRAM (Cache)
DRAM (RAM)
Speed ↓
Size ↑



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- Cache memory is a high-speed memory, which is small in size but faster than the main memory (RAM). The CPU can access it more quickly than the primary memory. So, it is used to synchronize with high-speed CPU and to improve its performance. Cache memory can only be accessed by CPU. It can be a reserved part of the main memory or a storage device outside the CPU. It holds the data and programs which are frequently used by the CPU.

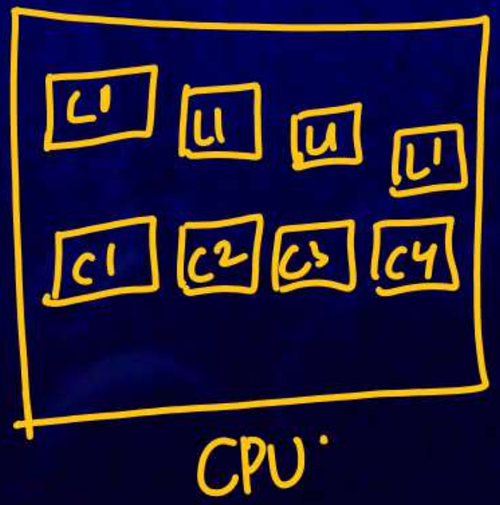
Topic : Memory Unit

- So, it makes sure that the data is instantly available for CPU whenever the CPU needs this data. In other words, if the CPU finds the required data or instructions in the cache memory, it doesn't need to access the primary memory (RAM). Thus, by acting as a buffer between RAM and CPU, it speeds up the system performance.

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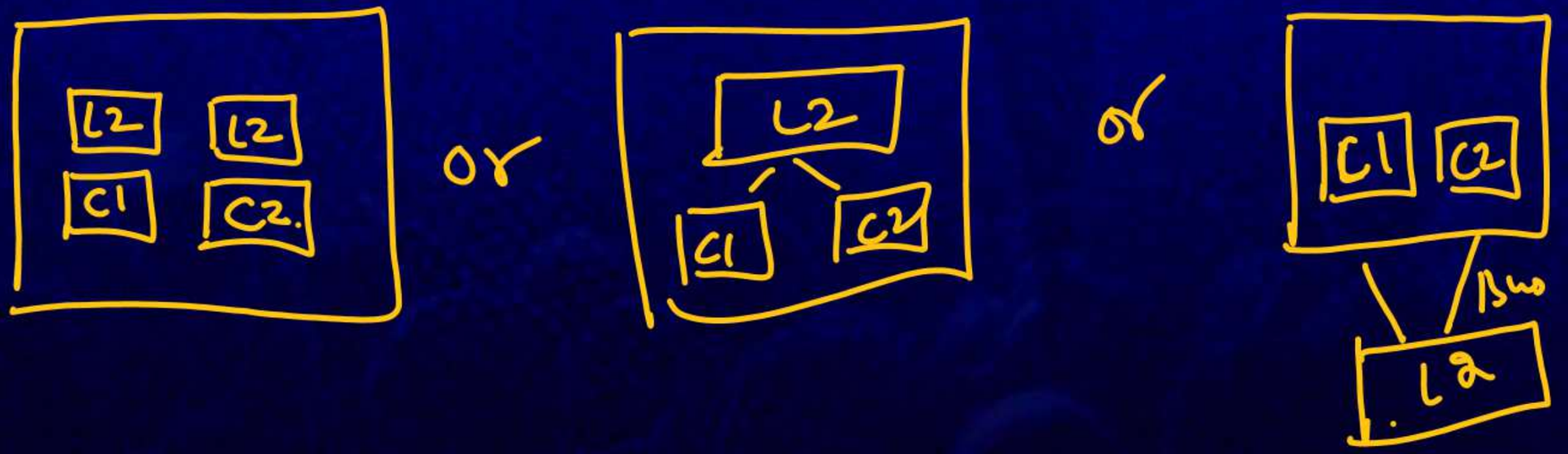
Types of Cache Memory:

- L1: It is the first level of cache memory, which is called Level 1 cache or L1 cache. In this type of cache memory, a small amount of memory is present inside the CPU itself. If a CPU has four cores (quad core cpu), then each core will have its own level 1 cache. As this memory is present in the CPU, it can work at the same speed as of the CPU.



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- L2: This cache is known as Level 2 cache or L2 cache. This level 2 cache may be inside the CPU or outside the CPU. All the cores of a CPU can have their own separate level 2 cache, or they can share one L2 cache among themselves. In case it is outside the CPU, it is connected with the CPU with a very high-speed bus.



Topic : Memory Unit

- L3: It is known as Level 3 cache or L3 cache. This cache is not present in all the processors; some high-end processors may have this type of cache. This cache is used to enhance the performance of Level 1 and Level 2 cache. It is located outside the CPU and is shared by all the cores of a CPU.

Topic : Memory Unit

Memory Hierarchy:

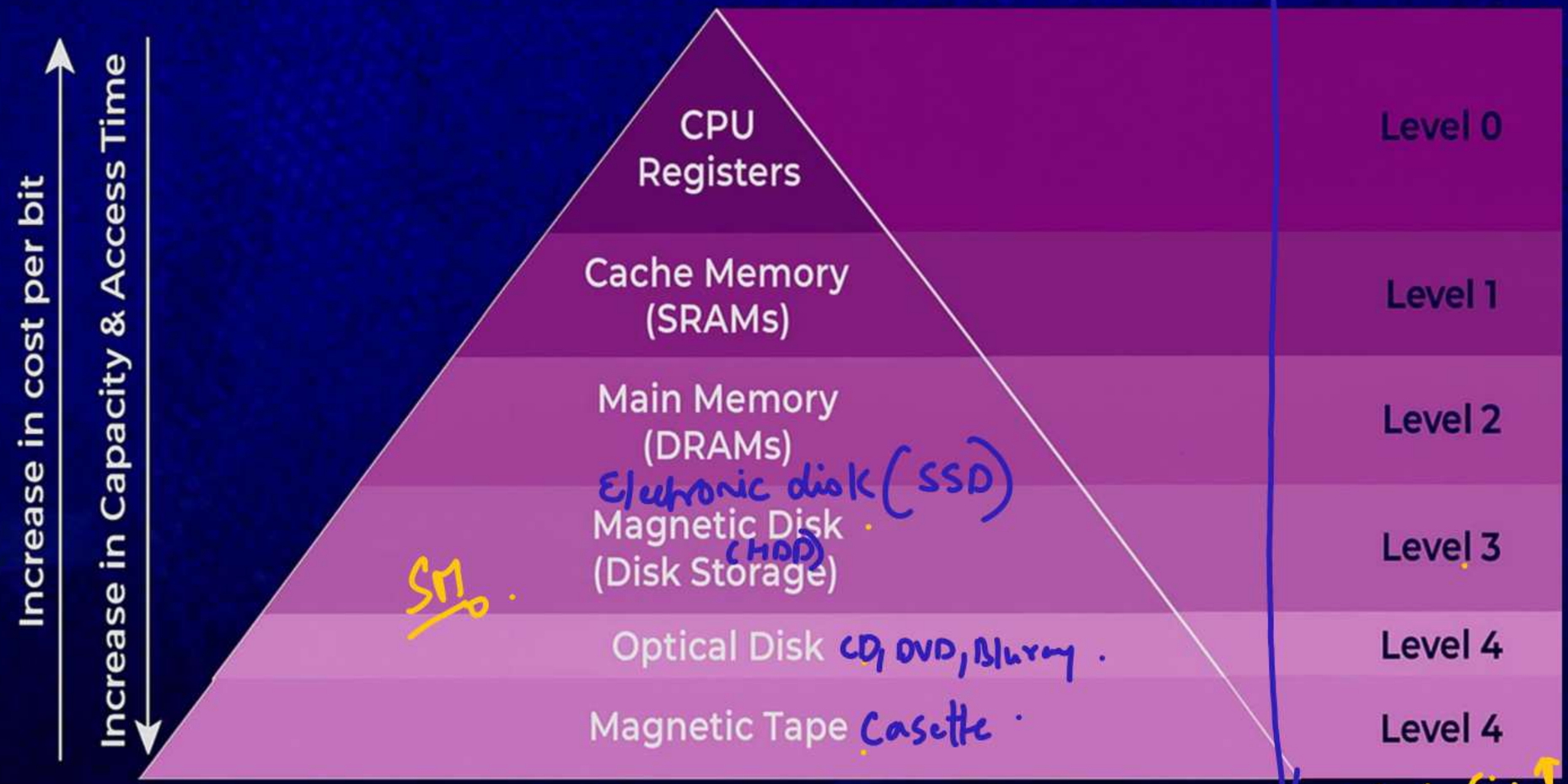
Memory Hierarchy Design is divided into 2 main types:

- External Memory or Secondary Memory: Comprising of Magnetic Disk, Optical Disk, and Magnetic Tape i.e. peripheral storage devices which are accessible by the processor via an I/O Module.
- Internal Memory or Primary Memory: Comprising of Main Memory, Cache Memory and CPU registers. This is directly accessible by the processor.

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V. imp.

LEARN.



Memory Hierarchy Design


(1) Arrange in ascending order acc. to access time.
(1) Registers. — (1)
(2) Cache — (2)
(3) HDD — (4)
(4) SSD. — (3)
(5) CD — (5)
least access time. ← CPU.
Registers
Cache.
RAM.
SSD
HDD.
Optical disk (CD).
Magnetic tape
Max Access time.
Access time ↑

Topic : Memory Unit

Memory unit:

Units of Computer Memory Measurement	
1	1 bit = Binary digit
2	8 bits = 1 Byte
3	1024 byte = 1 KB (Kilo Byte)
4	1024 KB = 1MB (Mega Byte)
5	1024MB = 1 GB (Giga Byte)
6	1024 GB = 1 TB (Tera Byte)
7	1024 TB = 1 Peta Byte
8	1024 PB = 1 Exa Byte
9	1024 EB = 1 Zetta Byte
10	1024ZB = 1 Yotta Byte
11	1024YB = 1 Bronto Byte
12	1024BB = 1 Geop Byte

2 Mins Summary

Memory unit



Homework from SAMARTH CUET PG 2027 Module

Rwise

+ DPP

Tmrw

Yt

Error detection
(Hamming distance, Error detection Methods).



THANK YOU